

Machine Learning Development Life Cycle (MLDLC)

Contents

1. Overview	3
2. From SDLC to MLDLC	4
2.1 The software development life cycle	4
2.2 The machine learning development life cycle	5
2.3 Why the whole cycle matters	5
2.4 The number of stages varies	5
3. Framing the problem	5
4. Gathering data	6
4.1 Data in a school project versus data in a company	6
4.2 Where data comes from	6
5. Data preprocessing	7
5.1 Why raw data cannot go straight into a model	7
5.2 Common preprocessing tasks	7
6. Exploratory data analysis (EDA)	8
6.1 What EDA is for	8
6.2 What we do during EDA	8
6.3 Time spent on EDA pays back later	8
7. Feature engineering and selection	9
7.1 Feature engineering	9
7.2 Feature selection	9
8. Model training, evaluation and selection	9
8.1 Training many algorithms	10
8.2 Evaluation with performance metrics	10
8.3 Model selection and hyperparameter tuning	11
8.4 Ensemble learning	11
9. Model deployment	12
9.1 From model to software	12
9.2 How a deployed model answers a request	12
10. Testing	13
10.1 Beta testing	13
10.2 A/B testing	13
10.3 When testing fails: going back	13

11. Optimizing	14
11.1 Launching at scale	14
11.2 Model drift and retraining	15
11.3 Cutting extra cost	15
12. The life cycle and the Pipeline map	15
13. Summary	18
Sources	18
14. Key terms	19

Where this fits: Pipeline map steps 0 (Foundations), 1 (Frame the problem), 2 (Get data), 3 (Understand data), 4 (Clean), 5 (Engineer features), 8 (Model), 10 (Tune), 11 (Deploy), 12 (Test), 13 (Monitor and maintain). See the [Course map](#).



- **Builds on:** Features ([Note 2](#)); Supervised learning ([Note 3](#)); Dimensionality reduction ([Note 3](#)); Batch (offline) learning ([Note 4](#)); Online learning ([Note 5](#)); Overfitting ([Note 7](#)).
- **Leads to:** ML pipelines ([Note 13](#)); Descriptive statistics ([Note 19](#)); Simple imputation (mean, median, mode, constant) ([Note 23](#)); Binning and binarization ([Note 23](#)); Feature transformation ([Note 23](#)); Feature extraction ([Note 23](#)).
- **Compare with:** Data mining ([Note 8](#)); JSON and SQL data ([Note 16](#)); Pandas Profiling ([Note 22](#)); Feature extraction ([Note 23](#)); Normalization ([Note 25](#)); Decision trees ([Note 97](#)).

1. Overview

Key point: Building an ML product takes nine stages, from framing the problem to keeping the live model healthy. Training a model is only one of them.

Figure 1 shows the nine stages and how they connect. Stage 1 happens first; stages 2 to 9 form a cycle, because a live model must be retrained on new data. When testing finds a problem, we go back to the stage that caused it.

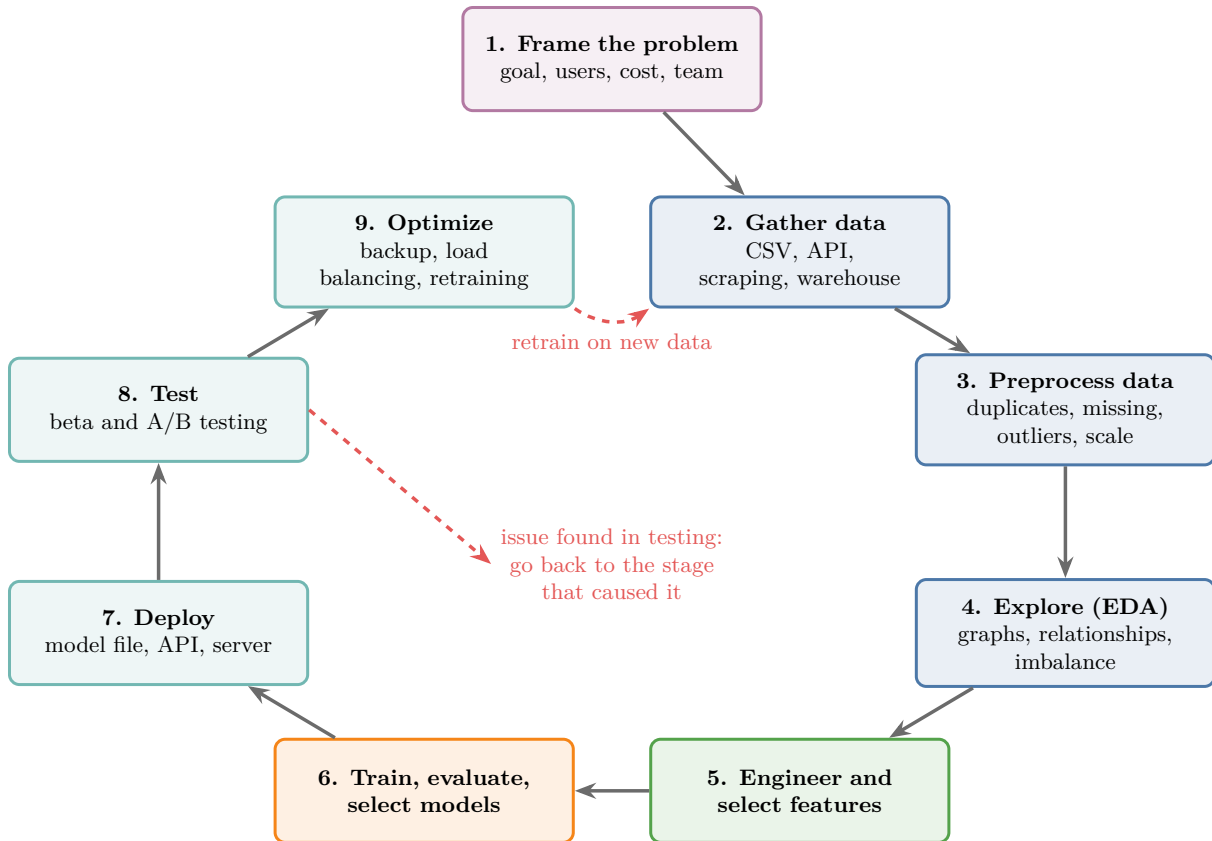


Figure 1: The nine stages of the ML development life cycle

Each stage gets its own Notes later. This Note gives the whole picture first, so that every later Note has a place on it.

2. From SDLC to MLDLC

Key point: The MLDLC is a set of guidelines for building an ML product from idea to finished product, in the same way the SDLC guides ordinary software.

2.1 The software development life cycle

Key point: Ordinary software is built by following a standard process, the SDLC.

The **software development life cycle (SDLC)** is the standard process for building a software product, from start to end. It is part of the subject *software engineering*. A software developer in a company follows it whenever they build a product.

2.2 The machine learning development life cycle

Key point: ML products are new, so a shared process for them was needed too.

Many companies now build software products that contain ML: a recommender system for a website, a loan-approval model for a bank, a model that predicts students' marks for a school. Each team used to work in its own way. Researchers wanted one common process that everyone could follow.

The result is the **machine learning development life cycle (MLDLC)**: a set of guidelines to follow whenever we build an ML-based software product. It guides us from the first idea to a working product.

2.3 Why the whole cycle matters

Key point: Training a model and checking its accuracy is not the end of the work. Companies want people who can build the whole product.

A common beginner's mistake is to train a model, look at its accuracy and stop there. In a company, the model must become a product that real users can use. Employers look for people with experience of building such **end-to-end** products, from raw idea to live software.

2.4 The number of stages varies

Key point: Different sources count 7, 9 or 10 stages, but the core idea is the same everywhere.

The MLDLC is not yet strictly defined, because ML is still a young field. One book may list fewer stages and another more. Here we use nine stages. If another source lists ten, it describes the same process, split differently.

3. Framing the problem

Key point: Before doing anything, we decide exactly what we are building, for whom, at what cost and how.

In a company, we work for customers, and every change of direction costs money. We cannot start, realise halfway that we planned the wrong thing, and start again. So the first stage is to **frame the problem**: answer the big questions before any work begins.

The questions we answer at this stage:

- What exactly is the problem, and what are we trying to solve?
- Who are the customers?
- How much will it cost?
- How many people does the team need?
- What will the end product look like?
- Is the ML problem supervised or unsupervised?

- Will the model learn in batch (offline) mode or online mode?
- Which kinds of algorithms might help?
- Where will the data come from?

Once these are answered, we have a clear mental plan of what comes next. Only then do we move to stage 2.

Extra: Supervised and unsupervised learning are explained in the Note on types of ML, and batch and online learning in their own two Notes. Framing a problem in depth, with a worked example, comes in the Note on framing an ML problem.

4. Gathering data

Key point: Without data there is no ML. In a company, the data is rarely ready-made, so we fetch it from wherever it lives and store it in a usable form.

4.1 Data in a school project versus data in a company

Key point: School and college projects come with data; company projects usually do not.

In school or college projects, data comes ready-made: from Kaggle, from course material, or downloaded from the internet. In a company, the data is often very specific to the business and not easily available. We have to fetch it ourselves.

4.2 Where data comes from

Key point: Five common sources: CSV files, APIs, web scraping, data warehouses and big-data clusters.

- **CSV files:** the easy case. We get the file and start working.
- **APIs:** we call an API from Python code, receive the data in JSON format, and convert it to our preferred format, usually a CSV file.
- **Web scraping:** the data is on a website but not offered for download. We write Python code that extracts it from the web pages. Hotel-comparison sites such as trivago collect hotel details from many hotel websites this way. Sites that compare a product's price across online shops also scrape those shops.
- **Databases and data warehouses:** the data is in the company's own database. We do not run ML directly on this live database, because a mistake could take the website down. Instead the data is copied into a separate **data warehouse** through a process called **ETL** (extract, transform, load), and we fetch our data from there.
- **Big-data clusters:** very large data is spread across many machines (a cluster), handled by tools such as Spark. We fetch the data from those clusters.

The goal of this stage is to fetch the data and store it in the right form, so that the work can start.

Extra: In ETL, *extract* copies data out of the source systems, *transform* cleans and reshapes it, and *load* writes it into the warehouse. Apache Spark is a widely used tool for processing data that is too big for one computer, by spreading the work over a cluster of machines (Zaharia et al. 2016). CSV files, JSON, SQL, APIs and web scraping each have their own Note in the data-gathering part of this series.

5. Data preprocessing

Key point: Data from outside is almost always dirty. Preprocessing turns it into a form the algorithm can use.

5.1 Why raw data cannot go straight into a model

Key point: Dirty data gives bad results, so it must be fixed first.

Data fetched from external sources is almost always **dirty** (unclean). If we pass it straight into a model, the results are poor. Typical problems:

- structural issues,
- missing data,
- outliers,
- noisy data,
- data from different sources that does not fit together, for example a different number of columns.

Data preprocessing means the changes we make to the data before the main processing (training).

5.2 Common preprocessing tasks

Key point: Remove duplicates, deal with missing values and outliers, and put columns on similar scales.

- **Remove duplicates:** rows that appear more than once.
- **Handle missing values:** empty cells.
- **Remove outliers:** values far from the rest.
- **Scale values:** bring columns to similar ranges.

Scaling matters because many algorithms compute distances between rows, and a column in crores would outweigh one in decimals (see Section 7 of the [toy project Note](#)). One common way to scale is standardization (see the [standardization Note](#)).

The core idea of the whole stage: bring the data into a format the ML algorithm can easily consume.

Extra: One crore is ten million (10,000,000). Removing rows is only one way to handle missing values; often we fill them in instead (imputation). Missing values, outliers and

scaling each have several Notes of their own in the feature engineering part of this series.

6. Exploratory data analysis (EDA)

Key point: Before building a model, we study the data closely: what each column looks like and how the inputs relate to the output.

6.1 What EDA is for

Key point: We cannot build a good model on data we do not understand.

Exploratory data analysis (EDA) is the stage where we analyse the data to find the relationships hidden in it, especially between the inputs and the output. We experiment with the data: plot graphs, look at numbers, test ideas. The aim is a concrete picture of the data in our mind, which makes every later decision easier.

6.2 What we do during EDA

Key point: Visualise the data, study columns one, two or several at a time, find outliers and check whether the classes are balanced.

- **Visualization:** plot graphs of the data.
- **Univariate analysis:** study each column on its own: its mean, its standard deviation, the shape of its distribution.
- **Bivariate analysis:** study two columns together, to see the relationship between them.
- **Multivariate analysis:** study three or four columns together.
- **Outlier detection:** find the values far from the rest.
- **Handling imbalanced data:** turn an imbalanced dataset into a balanced one.

A dataset is **imbalanced** when one class has far more rows than another. For example, in a dog-versus-cat image classifier, we might have many cat images and very few dog images. A model trained on such data tends to favour the large class (He and Garcia 2009, §2), so we handle the imbalance at this stage.

Extra: “Handling” imbalance usually means collecting more rows of the small class, creating extra copies or synthetic rows of it (oversampling), or dropping some rows of the large class (undersampling) (He and Garcia 2009, §3.1). Univariate, bivariate and multivariate analysis each have their own Note in the understanding-data part of this series.

6.3 Time spent on EDA pays back later

Key point: The more time we spend understanding the data, the less effort every later stage takes.

An old saying goes: if we have six hours to cut down a tree, we should spend four of them sharpening the axe. EDA is the sharpening. Experienced practitioners spend a lot of time

on it, because a good understanding of the data makes feature, model and testing decisions much easier.

7. Feature engineering and selection

Key point: Features are the input columns. We create better ones (feature engineering) and keep only the useful ones (feature selection).

The **features** are the input columns. The output depends on the inputs, so the quality of the features largely decides the quality of the model.

7.1 Feature engineering

Key point: We create new columns, or change existing ones intelligently, to make the data easier to learn from.

Feature engineering means creating new columns from the existing ones, or making intelligent changes to existing columns.

Example: house prices. Replacing the rooms and washrooms columns with one hand-made area column, as in Section 3.3 of the [types of ML Note](#), is feature construction (see Section 7 of the [feature engineering Note](#)).

Feature engineering is one of the most important techniques in the whole workflow, and it has several Notes of its own.

7.2 Feature selection

Key point: We drop features that do not affect the output, because they add nothing and slow training down.

Some datasets have 100 or 200 features. We do not keep them all, for two reasons:

1. **Many features do not help.** Not every input affects the output. We find the ones that do not and remove them.
2. **Fewer columns train faster.** The more columns there are, the longer training takes.

Choosing which features to keep is called feature selection; the [feature engineering Note](#) teaches it.

8. Model training, evaluation and selection

Key point: We train several algorithms, measure each with a performance metric, pick the best, tune it, and often combine several models into one.

Once the data is clean and the features are good, we are ready to train. Figure 2 shows the four steps of this stage.

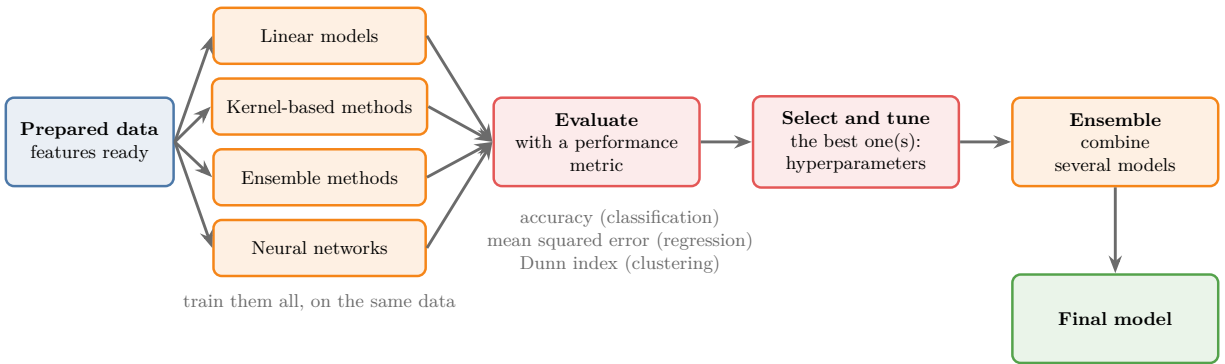


Figure 2: Train several algorithms, evaluate them, select and tune the best, then combine

8.1 Training many algorithms

Key point: We cannot know in advance which algorithm suits our data, so we train several and compare.

In practice, nobody trains just one algorithm. Some algorithms are known to suit some kinds of data: Naive Bayes, for example, performs very well on text. But another algorithm may do just as well or better on our particular data, and we only find out by trying.

So we train algorithms from different families on the same data:

- linear models,
- kernel-based methods,
- ensemble methods,
- neural networks.

Then we gather all the results to decide which model to use. **Model training** means giving the data to an algorithm so that it learns the pattern.

8.2 Evaluation with performance metrics

Key point: A performance metric is a number that tells us how well a model works. Each kind of problem has its own metrics.

In the **evaluation** step, we measure every trained model with **performance metrics**: numbers that tell us how well a model is working. They let us decide which model performs best. Common examples:

Problem	Example metric
Classification	Accuracy
Regression	Mean squared error
Clustering	Dunn index

Extra: *Accuracy* is the share of predictions that are correct. *Mean squared error* is the average of the squared differences between the predicted and the true values, so smaller is better. The *Dunn index* is the smallest distance between two clusters divided by the largest size (diameter) of any cluster, so it is higher when clusters are tight and far apart from each other (Dunn 1974). Each of these metrics, and many more, has its own Note later.

8.3 Model selection and hyperparameter tuning

Key point: We pick the best one or few algorithms and adjust their settings to get even better performance.

In **model selection**, we pick one or several of the best algorithms. Every algorithm has settings that we choose before training. Adjusting them to get the best performance is called **hyperparameter tuning**.

It is like adjusting a TV for a late-night movie: we change the picture mode and the sound mode and turn the volume up a little. Each setting is a knob; we turn the knobs until the picture is best for our situation.

Extra: In everyday speech these settings are often called “parameters”. Strictly, the settings we choose before training are **hyperparameters**, while **parameters** are the values the model learns from the data during training (for example, the slope of a line) (Goodfellow et al. 2016, §5.3). Tuning methods such as grid search and random search have their own Notes.

8.4 Ensemble learning

Key point: Combining several models into one usually gives a stronger model than any of them alone.

Ensemble learning connects several ML models to make one new, more powerful model. The main techniques are:

- bagging,
- boosting,
- stacking,
- cascading.

They differ in how the models are combined, but the core idea is the same: many models together make one strong model. Using an ensemble usually improves performance (Dietterich 2000), so it is a step we almost always take.

The full routine of this stage: train many models, evaluate them all, tune their hyperparameters, then apply ensemble learning. The result is one strong final model.

9. Model deployment

Key point: A trained model is not yet a product. We save it to a file, wrap it in an API, and put it on a server so users can reach it.

9.1 From model to software

Key point: Users never see the model; they see a website or an app that uses it.

Once we have a model that can make predictions, the main work is only beginning. We must turn it into software that people can use: a website, a mobile app or a desktop app. **Model deployment** means putting the model on a server so that it can answer users' requests.

9.2 How a deployed model answers a request

Key point: User input travels from the website to an API, the saved model makes a prediction, and the answer comes back as JSON.

Figure 3 shows the usual setup.

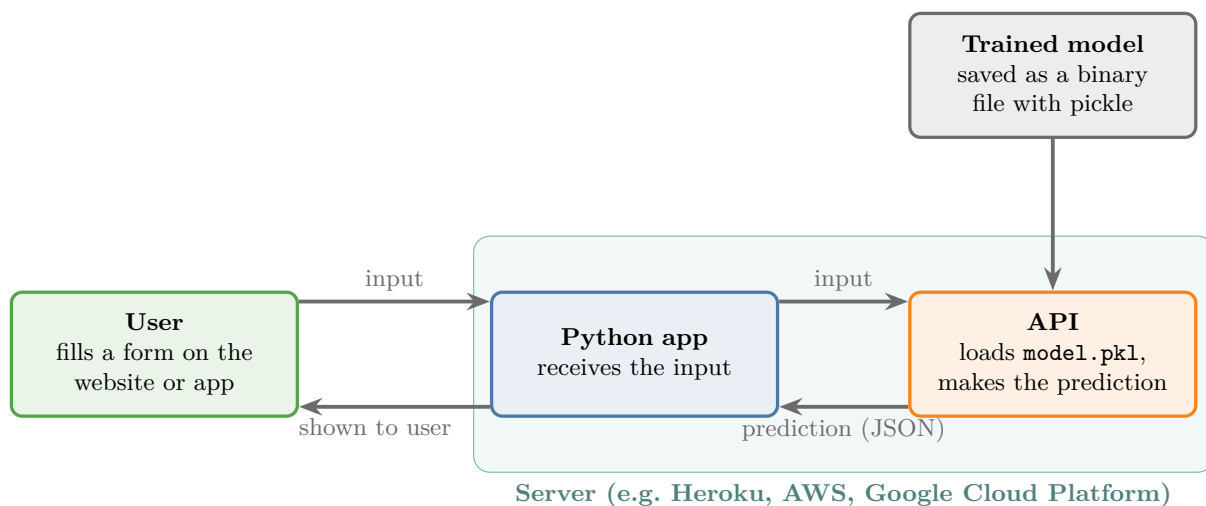


Figure 3: How a deployed model answers a user's request

1. **Save the model.** We save the trained model as a **binary file** (a file that is not plain text). A common tool for this in Python is **pickle**.
2. **Wrap it in an API.** An **API** here is a web address (URL) that, when given the right inputs, sends back an answer in JSON format. The API loads the saved model file.
3. **Answer a request.** The user enters values in a form on the website. The website passes them to a Python app on the server, which sends them to the API. The model makes a prediction, and the API returns it as JSON, which the app shows to the user.

The server itself is rented from a cloud provider such as Heroku, AWS (Amazon Web Services) or GCP (Google Cloud Platform). Once deployed, the model is online and serving users' requests. Later Notes build complete websites around ML models in this way.

Extra: **JSON** (JavaScript Object Notation) is a plain-text format for structured data, for example `{"prediction": "placed"}`. Heroku ended its free plan on 28 November 2022 (Heroku 2022); Render still has a free plan for small demos (Render docs). Pickle files should only be loaded from trusted sources, because loading a pickle can run code (Python docs, `pickle`).

10. Testing

Key point: We first release the new model to a small group of trusted users, compare it with the old version, and only then decide whether it works.

10.1 Beta testing

Key point: A new version goes to a small group of loyal users first, and their feedback decides what happens next.

Software updates rarely reach all users at once; they roll out gradually. **Beta testing** works the same way: we release the new model to a set of trusted, loyal customers who are likely to give good feedback, and collect that feedback.

10.2 A/B testing

Key point: A/B testing is a well-known way to check whether the new model really performs well.

A popular technique at this stage is **A/B testing**. After it, we decide whether the model we just built is working properly. A/B testing gets its own Note later.

Extra: In A/B testing, users are split at random into two groups. Group A keeps the current version and group B gets the new one. We then compare a measure that matters to the business, such as clicks or sales, between the two groups. Because the groups are random, a clear difference can be put down to the new version (Kohavi et al. 2020, Ch. 1).

10.3 When testing fails: going back

Key point: If the test goes badly, we return to the stage that caused the problem and redo the stages after it.

A problem can come from any earlier stage:

- the data was bad,

- preprocessing was not done properly,
- the feature selection was poor,
- the chosen algorithm has issues.

So we go back to that stage and redo the work from there. If the feedback is good, we move forward to the last stage. Figure 4 shows one pass through the cycle, a loop back after a failed test, and the retraining loop from section 11.

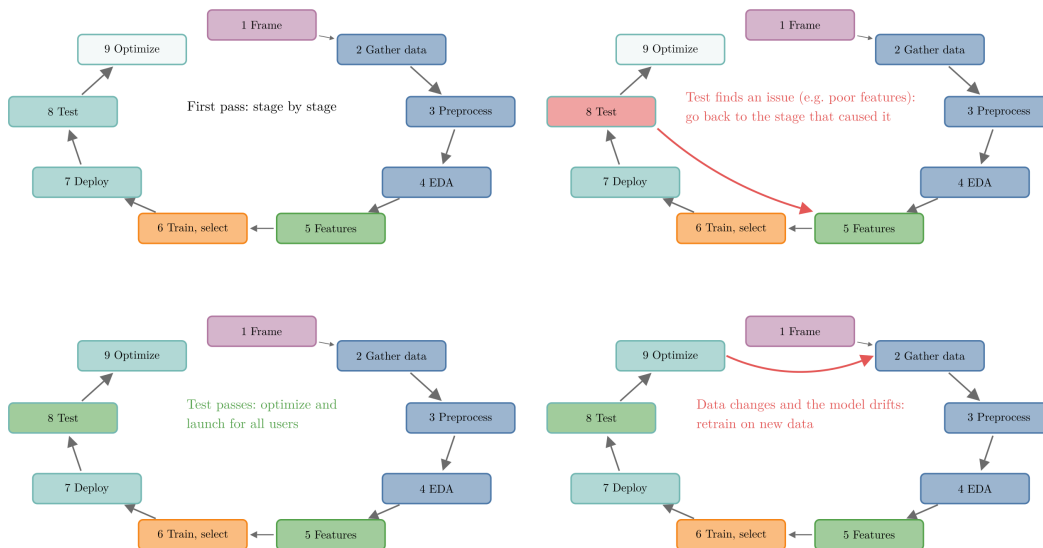


Figure 4: A failed test sends us back; a passed test leads to optimizing; a drifting model sends us back to new data

11. Optimizing

Key point: We launch the model for all users and set up everything that keeps it fast, safe and up to date: backups, rollback, load balancing and automatic retraining.

11.1 Launching at scale

Key point: Before the full launch, we protect the model and make sure the server can handle all users.

In the last stage, we launch the model on the server for all customers. Before that, we take several steps:

- **Back up the model and back up the data.**
- **Set up rollback:** if the live model breaks or something goes wrong, we automatically return to the last working version and put it live again.
- **Set up load balancing:** spread incoming requests across servers, so that many users at once are still served quickly.

11.2 Model drift and retraining

Key point: As the world changes, a model’s performance slowly gets worse, so we retrain it on new data at a fixed, automated schedule.

If a model is never retrained, its performance gets worse as the real-world data evolves away from its old training data. This is model drift, sometimes called model rot (see Section 4 of the [batch learning Note](#)).

Example: a mask detection system. It checks whether a person in front of a camera is wearing a mask. Then new kinds of masks appear, for example one whose lower half is printed to look exactly like a face. Our classifier will fail on these, so we need new data and must train the model again.

So we decide how often to retrain, for example weekly or monthly. This must be automated: we cannot repeat the whole process by hand every week.

11.3 Cutting extra cost

Key point: We look for every place where the process costs more than it needs to, and fix it.

Optimizing also means going through the whole process and removing extra expense wherever we find it, until the process runs smoothly.

Extra: Watching a live model’s performance so that we notice drift early is called **monitoring**. The practice of running and maintaining live models is called MLOps (Kreuzberger et al. 2023).

12. The life cycle and the Pipeline map

Key point: The Course map’s Pipeline map follows this life cycle but splits it into finer steps. Most stages match one step; a few differ.

The Pipeline map in the Course map uses 14 steps, based on these nine stages. Figure 5 and the table show how they line up.

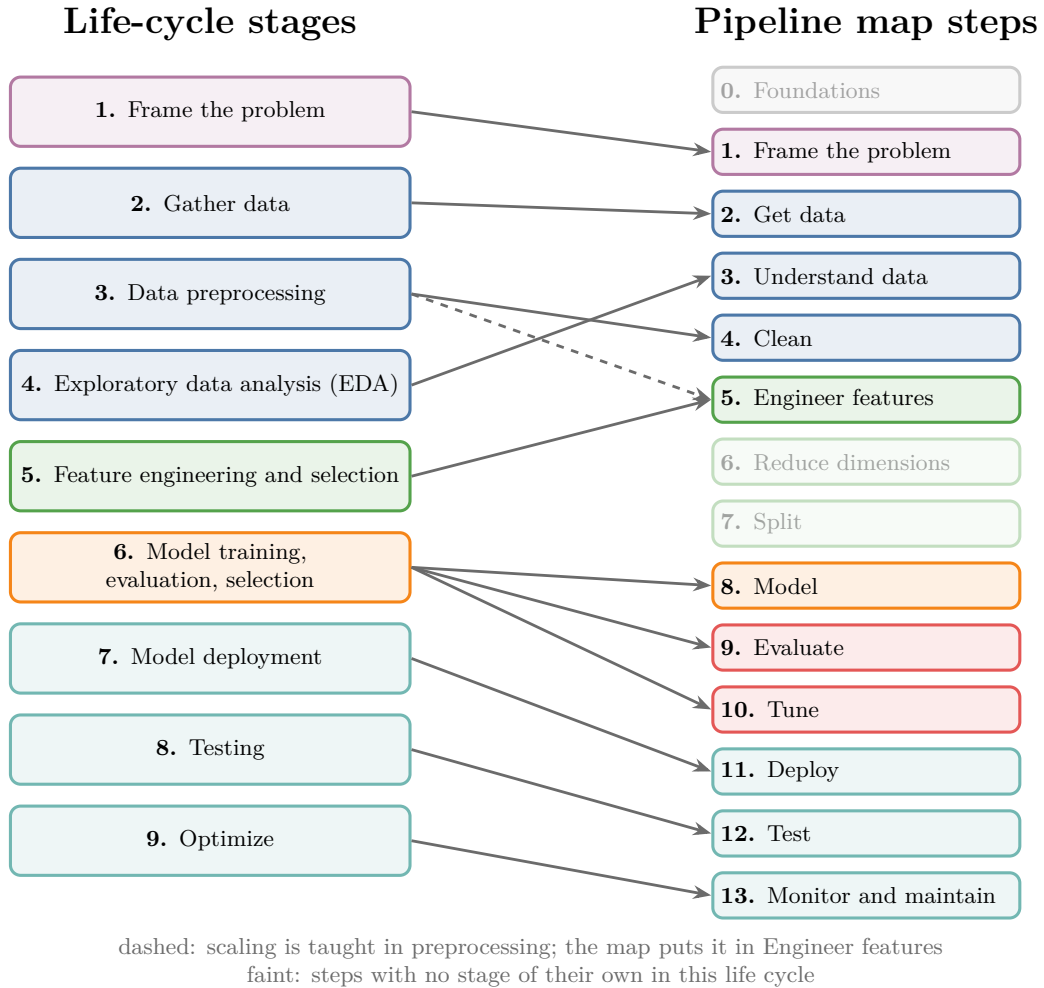


Figure 5: The nine life-cycle stages and the 14 Pipeline map steps

Life-cycle stage	Pipeline map step(s)
1. Framing the problem	1 Frame the problem
2. Gathering data	2 Get data
3. Data preprocessing	4 Clean (scaling: 5 Engineer features)
4. Exploratory data analysis	3 Understand data
5. Feature engineering and selection	5 Engineer features
6. Model training, evaluation and selection	8 Model, 9 Evaluate, 10 Tune
7. Model deployment	11 Deploy
8. Testing	12 Test
9. Optimizing	13 Monitor and maintain

Where the two differ:

- **Order of cleaning and EDA.** Here preprocessing comes before EDA. The Pipeline map puts Understand data (3) before Clean (4), with a loop arrow between them. Outlier detection appears in both stages here, which shows the same back-and-forth.
- **Scaling.** Here scaling is part of preprocessing. The Pipeline map puts it under Engineer features (5), with the other column transformations.
- **One stage, three steps.** Stage 6 covers training, evaluation and selection with tuning. The Pipeline map splits it into Model (8), Evaluate (9) and Tune (10). Ensemble learning sits in Model (8) on the map, because ensembles are algorithms in their own right.
- **Steps with no stage of their own.** Foundations (0), Reduce dimensions (6) and Split (7) are not stages here. Foundations holds background ideas rather than project work. Splitting the data into training and test sets, and reducing the number of dimensions, are taught in their own Notes.
- **Optimizing versus Monitor and maintain.** The optimizing stage covers backups, rollback, load balancing, retraining and cost. The map's step 13 groups the same work under a name that stresses watching the live model.

13. Summary

#	Stage	What we do	Main question
1	Framing the problem	Decide goal, customers, cost, team, type of ML, data source	What exactly are we building?
2	Gathering data	Fetch from CSV, API, scraping, warehouse (ETL), clusters	Where is the data?
3	Data preprocessing	Remove duplicates, handle missing values and outliers, scale	Can an algorithm use this data?
4	EDA	Visualise; univariate, bivariate, multivariate analysis; outliers; imbalance	What is in the data?
5	Feature engineering and selection	Create better columns; drop useless ones	Which inputs should the model see?
6	Training, evaluation, selection	Train many algorithms; compare with metrics; tune; ensemble	Which model is best?
7	Deployment	Save with pickle; wrap in an API; host on a server	How do users reach the model?
8	Testing	Beta testing, A/B testing; go back if it fails	Does it work for real users?
9	Optimizing	Backup, rollback, load balancing, scheduled retraining	How do we keep it healthy at scale?

- The MLDLC guides us **from idea to product**; training a model is only one stage of nine.
- Stage 1 comes first; stages 2 to 9 form a **cycle**, because live models drift and must be retrained on new data.
- A failed test sends us **back** to the stage that caused the problem.
- Sources count the stages differently; the core idea stays the same.

Sources

- Dietterich, T. (2000). Ensemble Methods in Machine Learning. *Multiple Classifier Systems*, LNCS 1857. Springer.
- Dunn, J. (1974). Well-Separated Clusters and Optimal Fuzzy Partitions. *Journal of Cybernetics* 4(1).
- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press.
- He, H. and Garcia, E. (2009). Learning from Imbalanced Data. *IEEE Transactions on Knowledge and Data Engineering* 21(9).
- Heroku (2022). Free Heroku Dynos, Heroku Postgres and Heroku Data for Redis are no longer available. Heroku Dev Center changelog.
- Kohavi, R., Tang, D. and Xu, Y. (2020). *Trustworthy Online Controlled Experiments*. Cambridge University Press.

- Kreuzberger, D., Kühl, N. and Hirschl, S. (2023). Machine Learning Operations (MLOps): Overview, Definition, and Architecture. *IEEE Access* 11.
- Python Software Foundation. `pickle`: Python object serialization. Python 3 documentation.
- Render. Deploy for Free. render.com/docs/free.
- Zaharia, M. et al. (2016). Apache Spark: A Unified Engine for Big Data Processing. *Communications of the ACM* 59(11).

14. Key terms

Term	Meaning
SDLC	Software development life cycle: the standard process for building ordinary software
MLDLC	Machine learning development life cycle: the guidelines for building an ML product from idea to product
End-to-end product	A complete product, from raw data to software that users use
Framing the problem	Deciding the goal, users, cost, team and approach before any work starts
Data warehouse	A separate store of copied company data, safe to analyse without touching the live database
ETL	Extract, transform, load: copying data from source systems into a warehouse
Dirty data	Data with errors, gaps, duplicates or inconsistencies
Data preprocessing	Changes made to the data before training, so an algorithm can use it
Exploratory data analysis (EDA)	Studying the data with graphs and summaries to find its patterns
Univariate analysis	Studying one column on its own
Bivariate analysis	Studying the relationship between two columns
Multivariate analysis	Studying three or more columns together
Imbalanced data	Data where one class has far more rows than another
Feature engineering	Creating new input columns, or changing existing ones, to help the model
Model training	Giving data to an algorithm so it learns the pattern
Performance metric	A number that measures how well a model works
Model selection	Choosing the best one or few algorithms after evaluation
Hyperparameter tuning	Adjusting an algorithm's settings for the best performance
Ensemble learning	Combining several models into one stronger model
Model deployment	Putting a model on a server so users can reach it
Binary file	A file that is not plain text, such as a saved model
Pickle	A Python tool for saving a model (or any object) to a file
JSON	A plain-text format for structured data, used by APIs

Term	Meaning
Beta testing	Releasing a new version to a small group of trusted users first
A/B testing	Comparing an old and a new version on two random groups of users
Rollback	Returning automatically to the last working version when something breaks
Load balancing	Spreading requests across servers so all users are served quickly